# SE LAB 5
# STATIC ANALYSIS REPORT

## 1)bandit_report.txt (before)

```
[main]   INFO  profile include tests: None
[main]   INFO  profile exclude tests: None
[main]   INFO  cli include tests: None
[main]   INFO  cli exclude tests: None
[main]   INFO  running on Python 3.13.1
Run started:2026-10-17 04:27:48.966995+00:00

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: ./ticket_booking_system.py:26:4
25         theater_seats[seat_id]["is_sold"] = False
26     except:
27         pass
28

--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: ./ticket_booking_system.py:42:18
41     base_price = theater_seats[seat_id]["price"]
42     final_price = eval(promo_expr)
43     return final_price

--------------------------------------------------
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: ./ticket_booking_system.py:47:11
46 def generate_ticket_hash(ticket_code):
47     return hashlib.md5(ticket_code.encode()).hexdigest()
48

--------------------------------------------------
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: ./ticket_booking_system.py:51:11
50 def generate_booking_pin():
51     return random.randint(1000, 9999)
52

--------------------------------------------------
>> Issue: [B605:start_process_with_a_shell] Starting a process with a shell, possible injection detected, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shell.html
   Location: ./ticket_booking_system.py:58:4
57     f.close()
58     os.system("echo Seating chart exported to: " + filename)
59

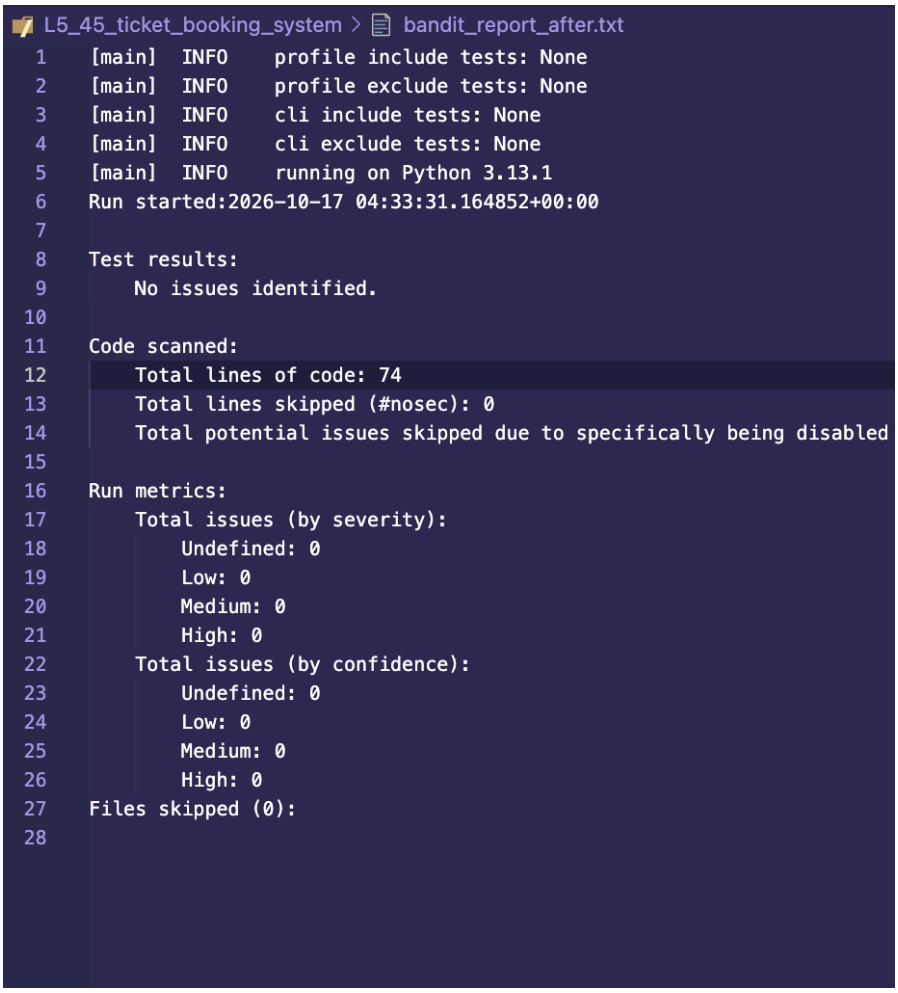
--------------------------------------------------

Code scanned:
   Total lines of code: 60
   Total lines skipped (#nosec): 0
   Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

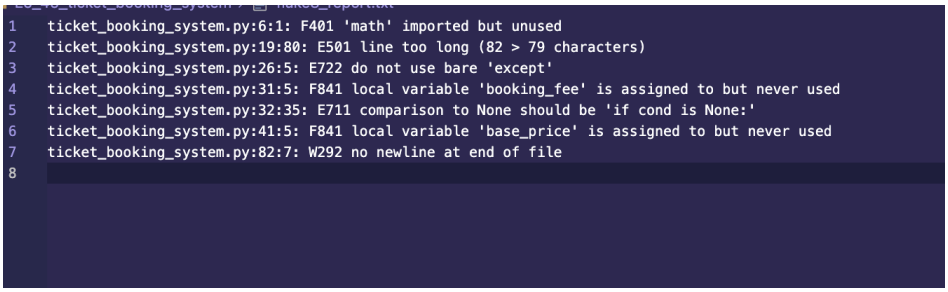
Run metrics:
   Total issues (by severity):
      Undefined: 0
      Low: 2
      Medium: 1
      High: 2
   Total issues (by confidence):
      Undefined: 0
      Low: 0
      Medium: 0
      High: 5
Files skipped (0):

```

bandit_report_after




2)
Flake8_report (before)




Flake8_report_after:



3)
Pylint_report (before):

```
************* Module ticket_booking_system
ticket_booking_system.py:82:0: C0304: Final newline missing (missing-final-newline)
ticket_booking_system.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:15:0: C0103: Function name "reserveSeat" doesn't conform to snake_case naming style (invalid-name)
ticket_booking_system.py:15:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
ticket_booking_system.py:19:25: C0209: Formatting a regular string which could be a f-string (consider-using-f-string)
ticket_booking_system.py:23:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:26:4: W0702: No exception type(s) specified (bare-except)
ticket_booking_system.py:30:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:32:7: C0121: Comparison 'theater_seats.get(seat_id) == None' should be 'theater_seats.get(seat_id) is None' (singleton-comparison)
ticket_booking_system.py:34:4: R1703: The if statement can be replaced with 'return bool(test)' (simplifiable-if-statement)
ticket_booking_system.py:34:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
ticket_booking_system.py:31:4: W0612: Unused variable 'booking_fee' (unused-variable)
ticket_booking_system.py:40:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:42:18: W0123: Use of eval (eval-used)
ticket_booking_system.py:41:4: W0612: Unused variable 'base_price' (unused-variable)
ticket_booking_system.py:46:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:50:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:54:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:55:4: C0103: Variable name "f" doesn't conform to snake_case naming style (invalid-name)
ticket_booking_system.py:55:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
ticket_booking_system.py:55:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
ticket_booking_system.py:61:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:61:24: W0621: Redefining name 'theater_seats' from outer scope (line 8) (redefined-outer-name)
ticket_booking_system.py:63:8: W0612: Unused variable 'seat' (unused-variable)
ticket_booking_system.py:69:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:6:0: W0611: Unused import math (unused-import)

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Your code has been rated at 5.19/10
```

Pylint_report_after:

```
--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 5.19/10, +4.81)
```